

AeroGuard TSLM

A multimodal time-series language model that gives turbofan engines a sense of time

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THE COST

\$150K+

per unscheduled engine removal (AOG) — delays, passenger accommodation, emergency charters.

LEGACY APPROACH

Fixed cycles

Calendar-based overhauls retire healthy parts early — or miss rapid wear from aggressive climb profiles.

PURE FORECASTING MODELS

Chronos

Univariate tokenization (Chronos) is blind to coupled aerothermal drift — and gives zero explanation.

TARGET USER

Aerospace Fleet Reliability Engineer & MRO Manager — needs an RUL number they can act on, and a reason they can trust.

From raw engine data to a plain-English answer

No engineering degree required to read the output — the AI does the interpreting.

1

Listen

Sensors on the engine record temperature, pressure and speed throughout every flight.

2

Learn

The AI studies 100 real jet engines, watched from healthy takeoff all the way to eventual failure.

3

Answer

For any engine, it reports how many flights are left — and explains why, in plain words.

WHAT YOU GET — THE NUMBER

“About 42 flights left”

Example answer — illustrative

WHAT YOU GET — THE REASON

“Running hotter and losing pressure in one section — a part is wearing out.”

Example answer — illustrative

51.1 RMSE on held-out engines — 85% lower NASA Score than the Chronos foundation model

Model	RUL RMSE↓	RUL MAE↓	NASA Score↓	Coverage
AeroGuard TSLM (ours)	51.11	39.81	2.62M	100% (806/806)
Chronos + Ridge (foundation model)	53.93	40.12	17.22M	100% (806/806)
Training mean (naive baseline)	73.43	59.09	129.27M	100% (806/806)
Text-only LM	—	—	—	0% — every response invalid

LIVE DEMO — MISSION CONTROL

- Interactive Plotly telemetry + digital-twin station cross-section
- Streaming Chain-of-Thought diagnostics as they generate
- ETOPS flight-route dispatch clearance recommendation

SAMPLE CoT OUTPUT (ILLUSTRATIVE) — UNIT #84

Observation: Ps30 falling, T50 surging → coupled HPC efficiency loss.
Reasoning: consistent with HPC blade erosion.
RUL: ~42 cycles remaining.
Action: inspect Station 30 HPC Rotor Blades (CFM56-HPC-RB25), Task Card AMM 72-31-00.

What we'd build with more runway

WHAT WORKED

- **Coupling beats univariate tokens** — reading Ps30 and T50 jointly, not separately, is what catches HPC drift.
- **Prefix fusion avoids tabular blindness** — the failure mode that makes text-only LLMs hallucinate part numbers.
- **A joint LM + RUL loss** keeps the diagnostic prose grounded in a real, checkable number.

LIMITATIONS

- **Single operating regime** — trained and evaluated on FD001 only, not yet FD002–004 or N-CMAPSS.
- **135M-parameter backbone** — sized for the hackathon's compute budget, not scaled up.
- **Dataset sourcing is template-driven** — not yet a fully autonomous agentic discovery pipeline.

NEXT STEPS

Multi-regime data

Extend to FD002–004 and N-CMAPSS

Agentic sourcing

Autonomous dataset discovery & validation

Scale the backbone

Larger LM + higher LoRA rank for richer CoT

Fleet integration

Wire ETOPS dispatch guidance into MRO ops

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